# A Mixing Thue–Morse Gap Shift with an Exact Zeta and a Natural Boundary

Route-A structural certificate C159

**Abstract**

We replace a one-pass symbolic interface by a recurrent renewal system. Let $S$ contain the indices at which the Thue–Morse bit is one, and require successive binary ones to be separated by a gap in $S$. The resulting S-gap shift is topologically mixing, has dense periodic points, and contains a recurrent transitive point. Unique circular renewal parsing gives its exact Artin–Mazur zeta. The Thue–Morse product then proves, by dense dyadic radial zeros, that the unit circle is a natural boundary for the source meromorphic continuation. This is a source theorem and not a target divisor, arithmetic factorization, or operator claim.

**Keywords:** symbolic dynamics; S-gap shift; Thue–Morse sequence; renewal system; Artin–Mazur zeta; topological mixing; natural boundary.

## 中文摘要

本文把一次性符号接口替换为可反复返回的更新动力系统。令集合 $S$ 由 Thue–Morse 序列中取值为一的下标组成，并规定相邻两个一之间的零间隙必须属于 $S$。 我们证明所得间隙移位拓扑混合、周期点稠密，且存在既递归又具有稠密轨道的点。 周期构型中每个一给出唯一圆周更新分解，因此可求得完整的 Artin–Mazur $\zeta$ 函数。进一步利用 Thue–Morse 无穷乘积在稠密二进单位根上的径向零，证明该源系统亚纯延拓的单位圆是自然边界。 该结论仅描述源动力系统，不构成目标除子、算术分解或算子实现。

关键词：符号动力系统；间隙移位；Thue–Morse 序列； 更新系统；Artin–Mazur $\zeta$ 函数；拓扑混合；自然边界。

## 1 A recurrent renewal shift

Write $t_s \in \{0,1\}$ for the parity of the binary digit sum of $s$ and put $S = \{s \geq 0 : t_s = 1\}$. Let $X_S \subset \{0,1\}^{\mathbb{Z}}$ consist of the all-zero point and the sequences in which successive ones have $s \in S$ zeros between them. Equivalently, its renewal code is $\mathcal{C} = \{10^s : s \in S\}$. One left shift is one clock tick.

Because $1, 2 \in S$, codeword lengths two and three are available. Any admissible word containing a one can first have its two partial end gaps completed to the nearest code boundaries; an all-zero word of length $L$ can be placed inside $10^{2^k}$ once $2^k \geq L$, since $t_{2^k} = 1$. After this boundary completion, words of lengths two and three fill every sufficiently large requested connector. Hence any two cylinders meet after every sufficiently large time: $X_S$ is topologically mixing.

Repeating a finite boundary-aligned code concatenation gives a periodic point in every cylinder containing a one. The cylinder $[0^L]$ is met separately by $(10^{2^k})^\infty$ for $2^k \geq L$, so periodic points are dense. To see recurrence constructively, enumerate the admissible words and recursively form a boundary-aligned block containing two copies of its predecessor and the next enumerated word, separated by two/three-length fillers. A centered limit contains every word and returns to each earlier nested cylinder. It is both recurrent and transitive. Thus recurrence is not inferred merely from the word “mixing.”

## 2 Exact zeta and entropy

Define, as formal series,

$$T(z) = \sum_{s \geq 0} t_s z^s, \quad F(z) = zT(z), \quad P(z) = \prod_{j \geq 0} (1 - z^{2^j}).$$

Unique binary expansion gives

$$P(z) = \sum_{s \geq 0} (-1)^{t_s} z^s = \frac{1}{1-z} - 2T(z). \tag{1}$$

Every nonzero periodic point has a unique circular parse at its ones. In $-\log(1-F) = \sum_{q \geq 1} F^q/q$, division by $q$ removes the $q$ choices of marked code boundary in a cyclic codeword list; converting boundary marks to the $n$

possible shift origins gives exactly the Artin–Mazur coefficient $\#\operatorname{Fix}(\sigma^n)/n$. Thus the renewal part contributes $(1-F)^{-1}$, while the all-zero fixed point contributes $(1-z)^{-1}$. Therefore

$$\zeta_{X_S}(z) = \frac{1}{(1-z)(1-F(z))} = \frac{2}{2-3z+z(1-z)P(z)}. \tag{2}$$

On the positive real axis $F$ is strictly increasing from zero to infinity. Its unique solution $F(R)=1$ gives $h_{\mathrm{top}} = -\log R$, and an exact rational partial sum plus geometric tail certifies

$$0.67633710444063914 < R < 0.67633710444063915. \tag{3}$$

## 3 The analytic boundary

If $\omega^{2^m}=1$ and $0<r<1$, the finite prefix of $P(r\omega)$ has modulus at most $2^m$; from level $m$ onward all factors are $1-r^{2^j}\in[0,1]$. Thus

$$|P(r\omega)| \le 2^m(1-r^{2^m}) \longrightarrow 0. \tag{4}$$

Dyadic roots are dense on $|z|=1$. Were $P$ meromorphic across an arc, the finite radial limit zero in (4) would exclude a pole at every dyadic root in that arc; each would be a zero (possibly after removing a removable singularity). Their interior accumulation would force the continuation to vanish identically, contradicting $P(0)=1$. Hence $P$ crosses no arc.

Equation (2) meromorphically continues the source zeta through the open unit disk. A continuation across any boundary arc, restricted to a subarc not containing 1, would also continue

$$P(z) = \frac{2/\zeta_{X_S}(z)-2+3z}{z(1-z)},$$

contradicting the preceding paragraph. The unit circle is therefore its natural boundary.

The period-18 and degree-48 ledgers are sentinels, not theorem cutoffs. An independent checker passes 747 assertions, SymPy passes 118 checks, canonical replay is byte-identical, and 46 hostile cases are rejected. The strict tuple is (`A1_WEAK`, `A2_FAIL`, `A3_PARTIAL_ANALYTIC_STRUCTURE`, `A4_FAIL`). The A3 label records only source analytic structure. We claim no target divisor, arithmetic/local factor, target functional equation or counting law, root number, automorphy, natural self-adjoint or Hilbert–Pólya lift, or Route-B authorization. Scope:

`NO_BAD_EULER_OR_ROOT_NUMBER`.

# Declarations

**Data and code availability.** Package-local evidence and code provide the exact certificate, independent checker, symbolic reconstruction, replay, and mutation audit.

**Ethics.** No human participants, animals, personal, clinical, or sensitive data are used; research ethics approval is not applicable.

**Author contributions (CRediT).** This anonymous certificate records Conceptualization, Formal analysis, Software, Validation, Writing—original draft, and Writing—review and editing at artifact level. AI systems are not authors.

**Conflicts of interest.** No conflict of interest is known.

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